

Exercise 1

Compute all the maximum and minimum points (global and local) of the function:

$$f(x) = 5 + 54x - 2x^3 \quad \text{on the interval } [0, 4]$$

1. The values of the function at the endpoints of the interval are: $f(0) = 5$ and $f(4) = 93$.

2. The derivative is:

$$f'(x) = 54 - 6x^2 = -6(x^2 - 9) = -6(x + 3)(x - 3)$$

We solve the equation:

$$f'(x) = 0 \iff -6(x + 3)(x - 3) = 0 \iff x = \pm 3$$

$$x_1 = 3 \in [0, 4] \quad \text{stationary point}$$

3. We have $f'(x) \geq 0$ for $x \in [-3, 3]$. Studying the sign of f' near $x = 3$, we deduce that $x = 3$ is a local maximum point and $f(3) = 113$ is a local maximum.

4. We have:

$$f(3) = 113 > f(0) = 5, \quad f(3) = 113 > f(4) = 93$$

We therefore conclude that:

- $f(0) = 5$ is the **global minimum** and $x = 0$ is a **global minimum point**
- $f(3) = 113$ is the **global maximum** and $x = 3$ is a **global maximum point**
- $f(4) = 93$ is a **local minimum** and $x = 4$ is a **local minimum point**

Exercise 2

Compute all the maximum and minimum points (global and local) of the function:

$$f(x) = 2x^3 - 3x^2 - 12x + 1 \quad \text{on the interval } [-2, 3]$$

1. The values of the function at the endpoints of the interval are: $f(-2) = -3$ and $f(3) = -8$.

2. The derivative is:

$$f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x+1)(x-2)$$

We solve the equation:

$$f'(x) = 0 \iff 6(x^2 - x - 2) = 0 \iff x = \frac{1 \pm 3}{2}$$

$$x_1 = -1 \in [-2, 3] \text{ and } x_2 = 2 \in [-2, 3] \text{ stationary points}$$

3. We have $f'(x) \geq 0$ for $x \in (-\infty, -1] \cup [2, +\infty)$. Studying the sign of f' near $x = -1$, we deduce that $x = -1$ is a local maximum point and $f(-1) = 8$ is a local maximum. Studying the sign of f' near $x = 2$, we deduce that $x = 2$ is a local minimum point and $f(2) = -19$ is a local minimum.

4. We have:

$$f(-1) = 8 > f(-2) = -3, \quad f(-1) = 8 > f(3) = -8$$

$$f(2) = -19 < f(-2) = -3, \quad f(2) = -19 < f(3) = -8$$

We therefore conclude that:

- $f(-2) = -3$ is a **local minimum** and $x = -2$ is a **local minimum point**
- $f(-1) = 8$ is the **global maximum** and $x = -1$ is a **global maximum point**
- $f(2) = -19$ is the **global minimum** and $x = 2$ is a **global minimum point**
- $f(3) = -8$ is a **local maximum** and $x = 3$ is a **local maximum point**

Exercise 3

Compute all the maximum and minimum points (global and local) of the function:

$$f(x) = x^3 - 6x^2 + 9x + 2 \quad \text{on the interval } [-1, 4]$$

1. The values of the function at the endpoints of the interval are: $f(-1) = -14$ and $f(4) = 6$.

2. The derivative is:

$$f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 3)(x - 1)$$

We solve the equation:

$$f'(x) = 0 \iff 3(x^2 - 4x + 3) = 0 \iff x = \frac{4 \pm 2}{2}$$

$$x_1 = 1 \in [-1, 4] \text{ and } x_2 = 3 \in [-1, 4] \text{ stationary points}$$

3. We have $f'(x) \geq 0$ for $x \in (-\infty, 1] \cup [3, +\infty)$. Studying the sign of f' near $x = 1$, we deduce that $x = 1$ is a local maximum point and $f(1) = 6$ is a local maximum. Studying the sign of f' near $x = 3$, we deduce that $x = 3$ is a local minimum point and $f(3) = 2$ is a local minimum.

4. We have:

$$f(1) = 6 > f(-1) = -14, \quad f(1) = 6 \geq f(4) = 6$$

$$f(3) = 2 > f(-1) = -14, \quad f(3) = 2 < f(4) = 6$$

We therefore conclude that:

- $f(-1) = -14$ is the **global minimum** and $x = -1$ is a **global minimum point**
- $f(1) = 6$ is the **global maximum** and $x = 1$ is a **global maximum point**
- $f(3) = 2$ is a **local minimum** and $x = 3$ is a **local minimum point**
- $f(4) = 6$ is the **global maximum** and $x = 4$ is a **global maximum point**

Exercise 4

Compute all the maximum and minimum points (global and local) of the function:

$$f(x) = x^4 - 2x^2 + 3 \quad \text{on the interval } [-2, 3]$$

1. The values of the function at the endpoints of the interval are: $f(-2) = 11$ and $f(3) = 66$.

2. The derivative is:

$$f'(x) = 4x^3 - 4x = 4x(x^2 - 1)$$

We solve the equation:

$$f'(x) = 0 \iff 4x(x^2 - 1) = 0 \iff x = \pm 1 \text{ and } x = 0$$

$$x_1 = -1 \in [-2, 3], \quad x_2 = 0 \in [-2, 3] \text{ and } x_3 = 1 \in [-2, 3] \text{ stationary points}$$

3. We have $f'(x) \geq 0$ for $x \in [-1, 0] \cup [1, +\infty)$. Studying the sign of f' near $x = -1$, we deduce that $x = -1$ is a local minimum point and $f(-1) = 2$ is a local minimum. Studying the sign of f' near $x = 0$, we deduce that $x = 0$ is a local maximum point and $f(0) = 3$ is a local maximum. Studying the sign of f' near $x = 1$, we deduce that $x = 1$ is a local minimum point and $f(1) = 2$ is a local minimum.

4. We have:

$$f(-1) = 2 < f(-2) = 11, \quad f(-1) = 2 < f(3) = 66$$

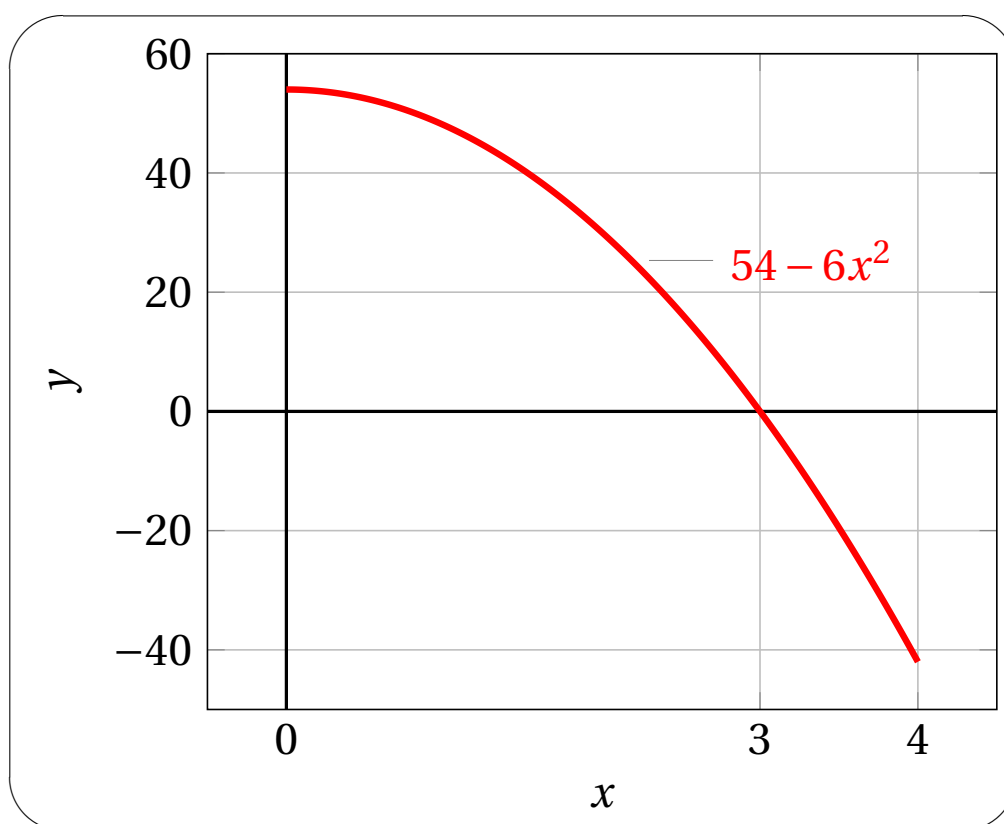
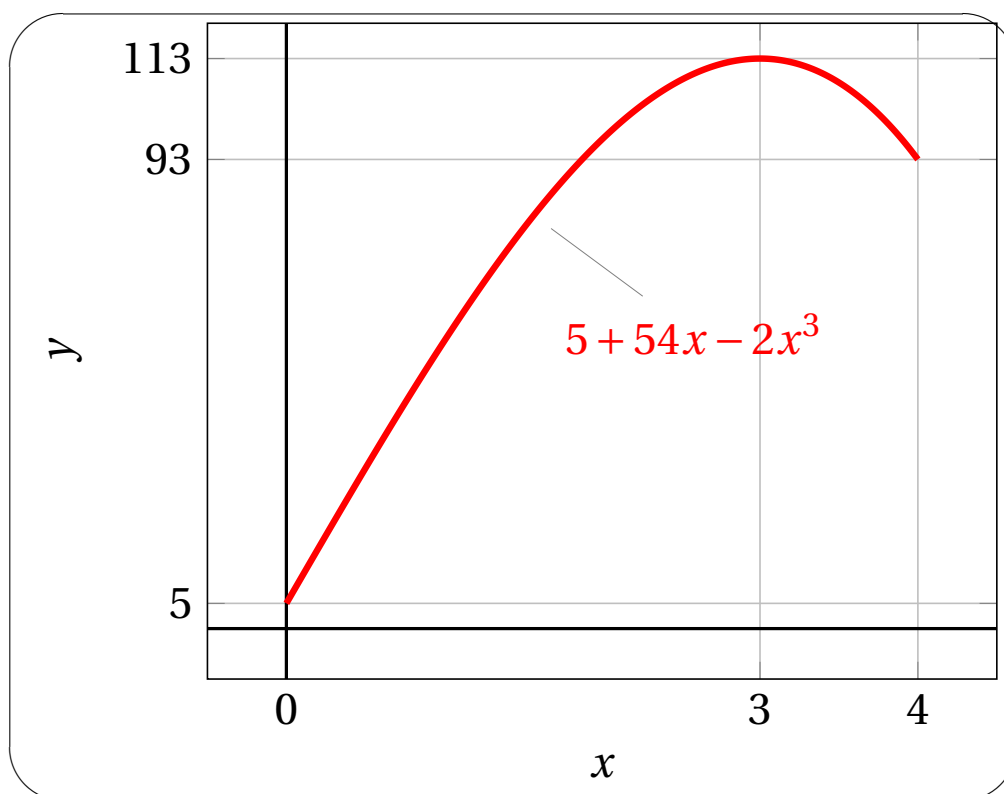
$$f(0) = 3 < f(-2) = 11, \quad f(0) = 3 < f(3) = 66$$

$$f(1) = 2 < f(-2) = 11, \quad f(1) = 2 < f(3) = 66$$

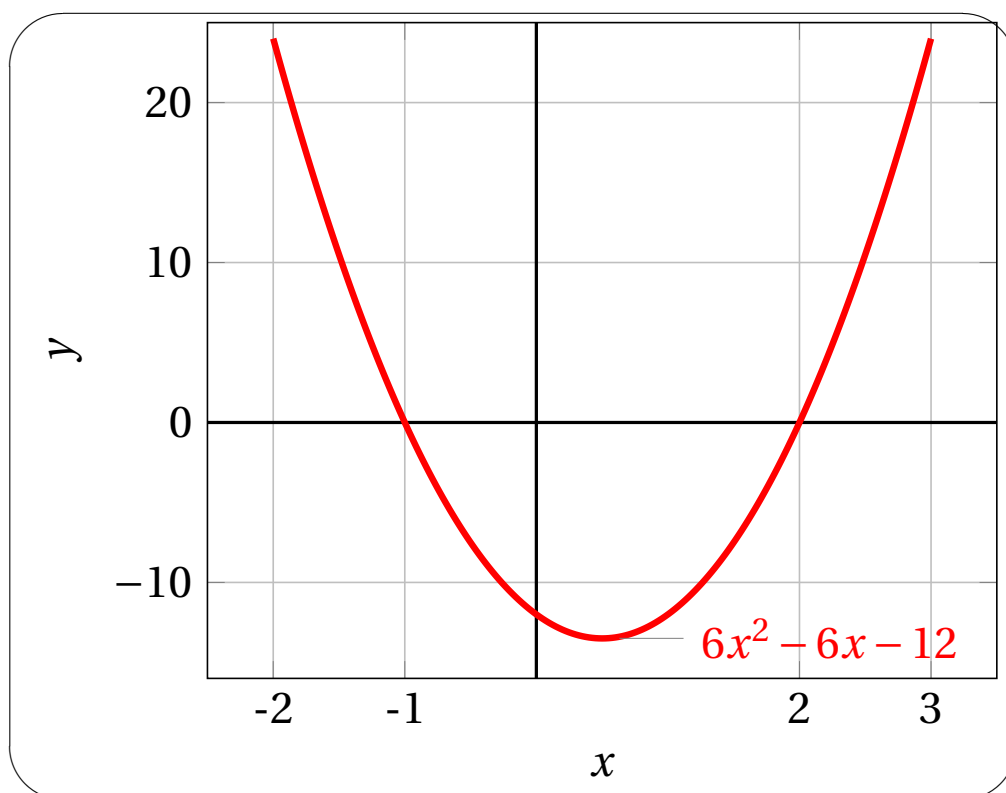
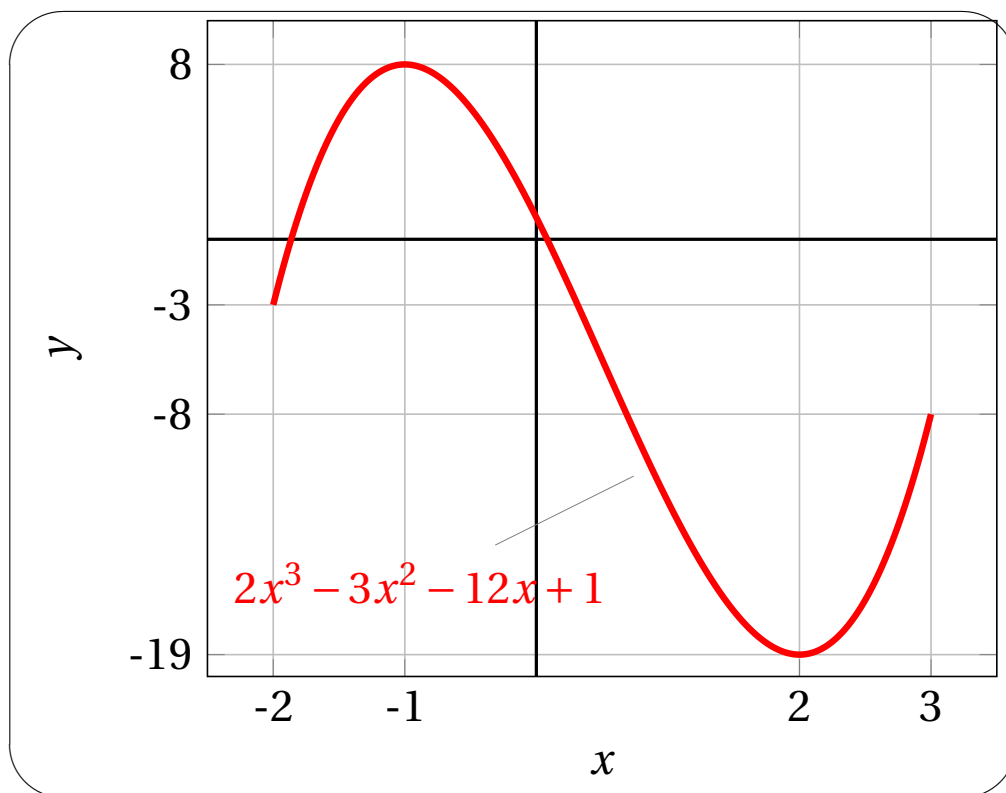
We therefore conclude that:

- $f(-2) = 11$ is a **local maximum** and $x = -2$ is a **local maximum point**
- $f(-1) = 2$ is the **global minimum** and $x = -1$ is a **global minimum point**
- $f(0) = 3$ is a **local maximum** and $x = 0$ is a **local maximum point**
- $f(1) = 2$ is the **global minimum** and $x = 1$ is a **global minimum point**
- $f(3) = 66$ is the **global maximum** and $x = 3$ is a **global maximum point**

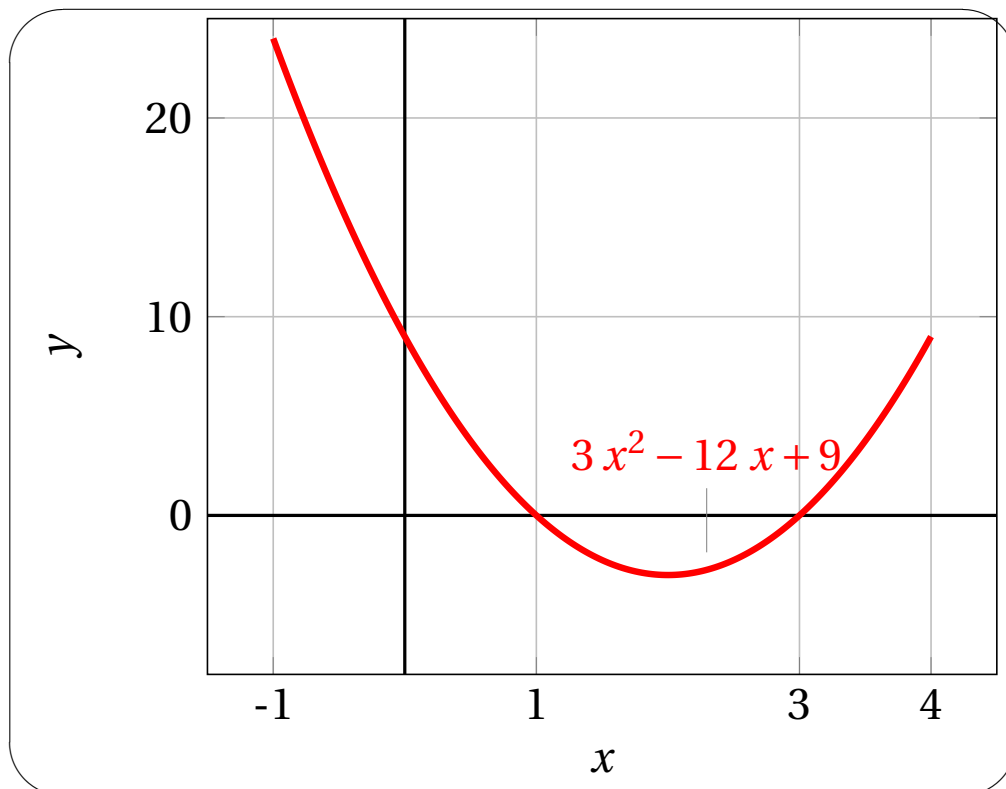
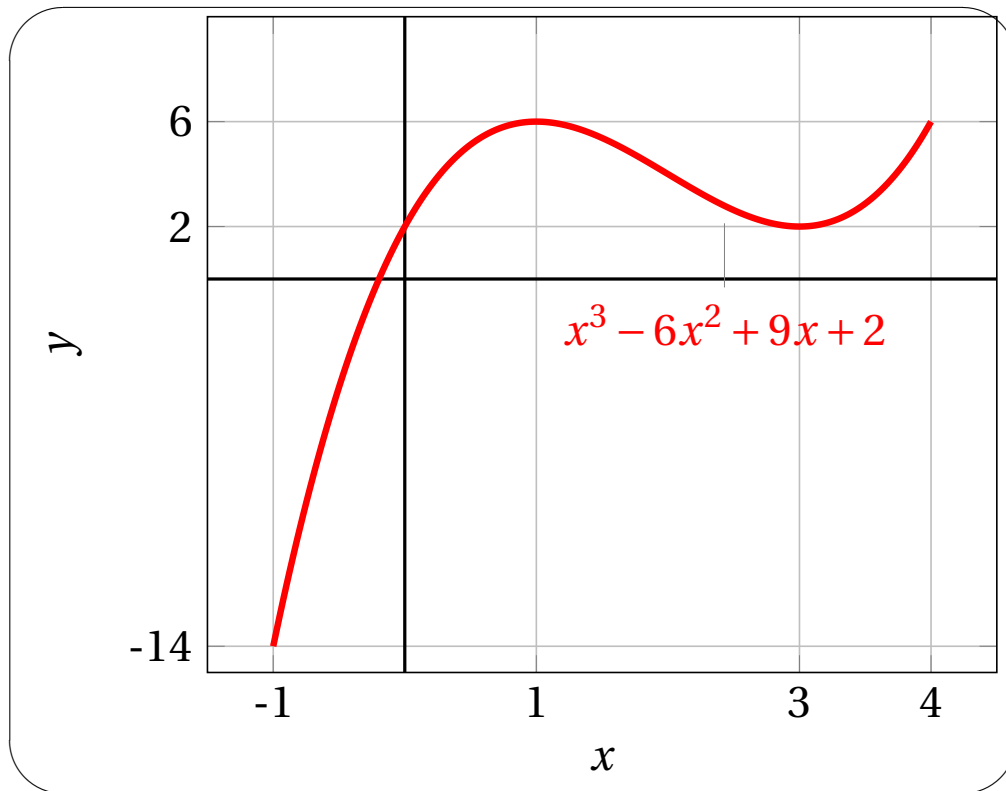
These are the graphs of the function $f(x) = 5 + 54x - 2x^3$ and of its derivative on the interval $[0, 4]$:



These are the graphs of the function $f(x) = 2x^3 - 3x^2 - 12x + 1$ and of its derivative on the interval $[-2, 3]$:



These are the graphs of the function $f(x) = x^3 - 6x^2 + 9x + 2$ and of its derivative on the interval $[-1, 4]$:



These are the graphs of the function $f(x) = x^4 - 2x^2 + 3$ and of its derivative on the interval $[-2, 3]$:

